COLOSSAL-A-PITCH HACKATHON

SMART CAMPUS MANAGEMENT & STUDENT SECURITY

HOSTEL 360

SMART HOSTEL ALLOCATION & OPTIMIZATION SYSTEM

Video link : https://drive.google.com/file/d/1EjR0Ok0Q12hZAbYfyOkXM90oLX6zIhQM/viewusp=drive_link

Rule-Driven, Zero-Cost, Transparent Hostel Allocation ERP

"Eliminating Discretionary Bias, Quota Miscalculations, and Spreadsheet Chaos in University Hostel Allotments"

Youtube link: https://youtu.be/I2alxETkmbw?si=hKQuHgNFtjlRb4x2

TEAM KRAKEN THE CODE

Github Link : https://github.com/SumitInTech/Hostel360.git

Team member : Sumit , Kunal, Vaibhav, Ankit

HOSTEL 360

THE IDEA

Hostel allocation is often a first-come-first-served, fragmented and manual process, requiring students to visit the institution for applications, verification, counselling, fee payment and registration.

This is especially challenging for students from distant locations, while genuine need, accessibility and room availability are difficult to consider together.

Hostel 360 brings the complete hostel journey online, enabling students to apply, complete verification, make payments and register from home through a centralized platform.

HOW IT ADDRESSES THE CHALLENGE

DIGITIZES THE HOSTEL PROCESS

Physical visits → Online portal
Students apply, upload documents and submit preferences directly from home.

VERIFIES & ESTABLISHES PRIORITY

Self-declared data → Verified need
Address/documents are verified → Routes API calculates distance → priority is generated based on distance and eligibility.

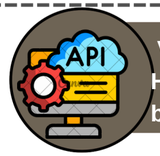
OPTIMIZES ROOM ALLOCATION

First-come-first-served → Need-based matching
Allocation engine applies eligibility, accessibility, institutional rules and room availability to assign the best available single-bed room.

COMPLETES REGISTRATION DIGITALLY

Multiple offices → One platform
Student views allocation → attends counselling if required → pays online → completes registration → receives digital allotment letter

UNIQUE FEATURES/ INNOVATIONS



Verified Distance Priority
Hometown-to-campus distance based ranking



Digital Document submission and verification



Smart Room Matching and Optimized allocation



Rule-Driven Allocation, without using probabilistic AI



Zero-Cost System, Serverless, and No Need for DevOps



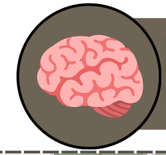
Online payment and DD verification after room confirmation



Smart Waitlist Mechanism



Live Admin Dashboard
Occupancy, vacancies & application insights



SMRITI Agent Memory
Recalls past warden precedents & overrides

TECHNICAL APPROACH

TECH STACK

Frontend	HTML • CSS • JavaScript -> Student & Admin Web Portal
Backend	Google Apps Script V8 -> Business Logic • Verification • Allocation Engine
Database	Google SpreadSheets
Document Management	Google Drive + Google Docs -> Documents • Allotment Letters • Audit Reports
Distance Calculation	Google Routes API -> Precise Real-World distance calculation from the Institute
Agent Memory	Hindsight Cloud Engine -> Retain · Recall · Reflect via REST API
Deployment	Apps Script Web App -> Serverless • Low-Cost • Easy Deployment

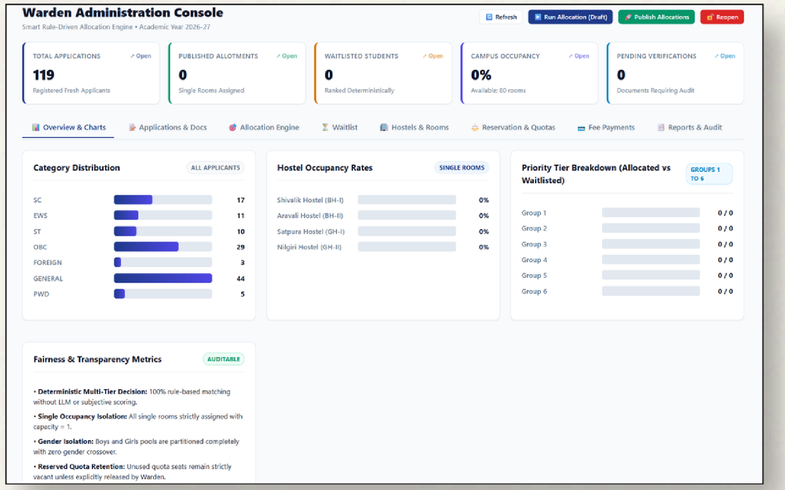
WORKFLOW

STUDENT INPUT

1. Hostel application
2. Hometown/address
3. Preferences & accessibility requirements
4. Valid Documents upload

DOCUMENT VERIFICATION

Digital verification of documents through Warden side



ALLOCATION ENGINE

Initial Hard Constraints

1. Single-room Occupancy
2. Gender Partition

6-tier Priority Hierarchy

1. PwD / Specially-abled students
2. Outside Delhi Residents
3. Delhi Outstation Residents
4. Exempted Residents (parents posted outside Delhi)
5. Local Delhi-NCR Residents
6. General and Late Applicants

Category Quotas for reserved seats

General (40.5%), OBC (27%), SC (15%), ST (7.5%), EWS (10%), PwD (5%), Foreign (5%)

Preference Engine

Preference 1 (if available) -> Preference 2 (if available) -> Official Waitlist (if both preferences are unavailable)

DRAFT ALLOCATION

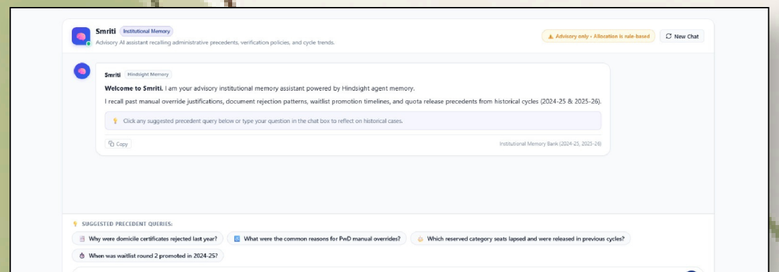
The draft is sent to the Warden

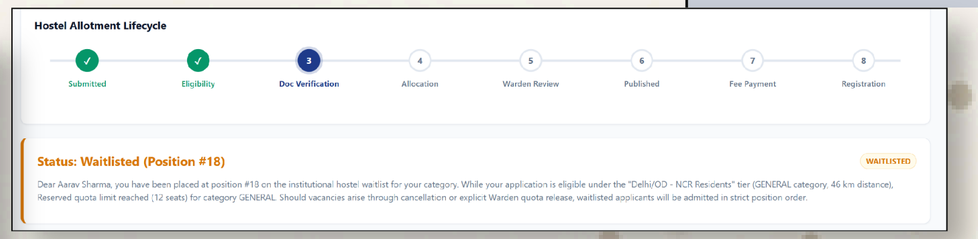
ADMIN VALIDATES

Admin validates the alloted hostel

PUBLISHED

Email is sent to the student and the alloted hostel is published





SMRITI: Institutional Memory for the Hostel Office

Powered by Hindsight Agent Memory (Retain · Recall · Reflect)

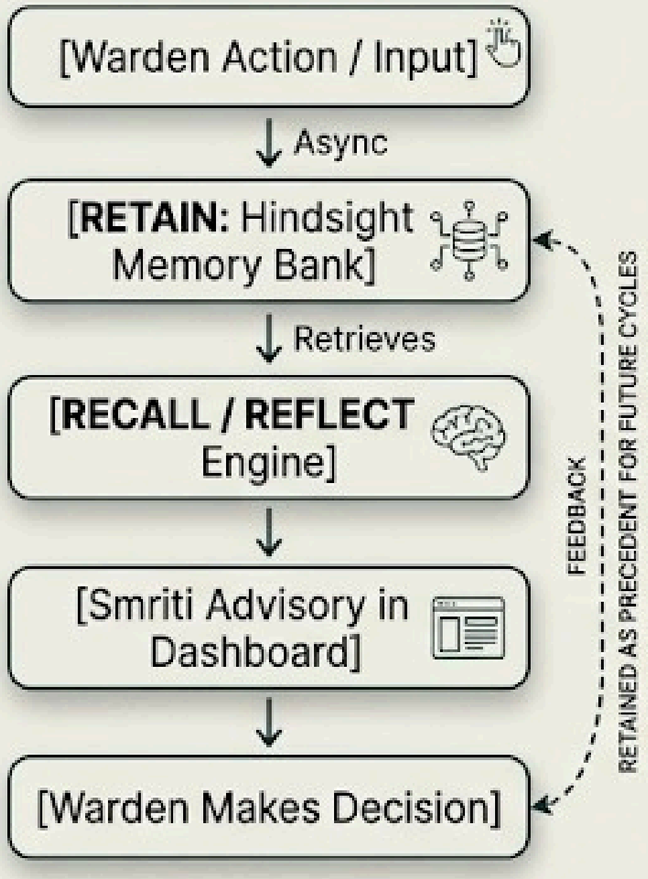
1. THE PROBLEM

- **Annual turnover** of wardens and clerks causes institutional amnesia.
- **Historical rationale** for overrides & rejected docs is lost.
- Every cycle starts from zero, repeating past disputes.

WHAT SMRITI DOES:

- **Retains:** Overrides, rejections, & cycle summaries.
- **Recalls:** Past precedents right during warden decisions.
- **Reflects:** Pre-publish briefings on multi-year trends.
- **Copilot Chat:** Natural language Q&A over historical cycles.

2. SYSTEM WORKFLOW



3. EVOLUTION ACROSS CYCLES

- **CYCLE 1 (Cold Start):** Zero history; baseline generic rule enforcement.
- **CYCLE 2 (Emerging Patterns):** Correlates recurring document ambiguities & payment delays.
- **CYCLE 3 (Mature Memory):** Recalls specific historical cases, CMO medical precedents, and quota interpretations.

⚠ **KEY DESIGN RULE:** 100% Advisory Only. Allocation stays 100% deterministic; 12-pt validator remains untouched.

FEASIBILITY AND VIABILITY

FEASIBILITY



Technical:

Built using the Google Workspace ecosystem — Apps Script, Sheets, Drive and HTML/CSS/JS — enabling a low-cost, serverless and easily deployable solution.



Operational:

Digitizes the complete hostel workflow from application and verification to allocation, payment and registration, reducing manual workload.



Economic:

Uses existing Google infrastructure with minimal deployment and maintenance costs, making it suitable for resource-constrained institutions.

CHALLENGES

Data Reliability:

Incorrect addresses, duplicate applications and incomplete documents can affect fair allocation.

User Adoption:

Government or address documents require secure handling, controlled access and reliable verification.

Accessibility:

Allocation rules differ across the institutions, requiring configurable policies and staff acceptance.

Knowledge Loss:

Annual warden rotation erases discretionary precedents & override history.

SOLUTIONS

Smart Validation: Automated checks for duplicates, missing information and address inconsistencies, with manual verification for flagged cases.

Secure Data Handling: Role-based access, limited data collection, audit logs and controlled document access.

Configurable Allocation: Administrators can define eligibility, priorities, accessibility rules, weights and deadlines without changing the core system.

Persistent Memory: SMRITI retains anonymized audit history across academic cycles.

IMPACT AND BENEFITS

OUR IMPACT ON:

Students

Fairer, transparent and need-based hostel allocation with a digital admission experience.

Hostel Administration

Reduced paperwork and faster room allocation, plus zero memory loss: warden decisions backed by past institutional precedents.

Institution

Better utilization of hostel capacity with centralized records, analytics and accountable decision- making.

WHY CHOOSE Hostel 360?

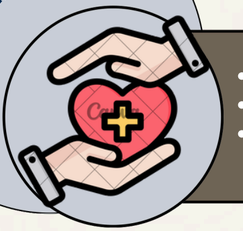
Fair & Need-Based
Moves beyond first-come-first-served allocation.

End-to-End Digital
Apply → Verify → Allocate → Pay → Register

Verified & Transparent
Address verification + explainable allocation decisions.

Institutional Memory
Never resets to zero; remembers multi-year administrative context & precedents.

BENEFITS



SOCIAL

- Fair access based on Need and Distance
- Improved accessibility for PwD Students
- Transparent allocation and reduced bias



ECONOMIC

- Low-cost deployment using Google Workspace
- Reduced administrative paperwork
- Better utilization of available hostel capacity



DIGITAL

- Digital application, verification and registration
- Digital documents, receipts and allotment letter generation

RESEARCH AND REFERENCES

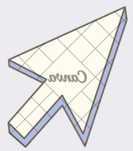
RESEARCH PAPERS

- Students' Accommodation Allocation: A Multicriteria Decision Support System — AHP, Weighted Sum & PROMETHEE for student accommodation ranking.
- Stable Matching of Student-Groups to Dormitories — Fair and stable approaches to student–dormitory allocation.
- Multi-Level Genetic Algorithm for Hostel Space Allocation — Optimization approaches for real-world university hostel allocation.

TECHNICAL REFERENCES

- Google Apps Script Documentation — Web apps, automation and Google Workspace integration. Google Apps Script Services & APIs — Sheets, Drive, Gmail and external API integration.
- Google Apps Script Quotas & Limitations — Deployment and scalability considerations.
- Hindsight Agent Memory (Vectorize) — Retain · Recall · Reflect architecture.

RESEARCH INSIGHTS USED



- Multi-criteria ranking → Priority-based student allocation
- Stable/fair matching → Transparent room assignment
- Optimization techniques → Efficient use of limited rooms
- Data protection principles → Secure document & address handling
- Episodic Agent Memory → Explainable, precedent-backed administrative reasoning.